

1 UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

2 DOCTORAL THESIS

3

4 **The study of Higgs properties in the**
5 **four-lepton final state at CMS**

6

7 *Author:*

Tahir JAVAID

Supervisor:

Prof. Guoming CHEN

Co-supervisor:

Prof. Mingshui CHEN

8 *A thesis submitted in fulfillment of the requirements*
9 *for the degree of Doctor of Science*

10 *in the*

11 Institute of High Energy Physics
12 Chinese Academy of Sciences



14 **Contents**

15 **List of Figures** **v**

16 **List of Tables** **vii**

17	1 Summary and Outlook	1
18	1.1 Summary of the dissertation	1
19	1.2 Outlook	2

²⁰ List of Figures

²¹ **List of Tables**

Chapter 1

Summary and Outlook

1.1 Summary of the dissertation

After many years of search, ATLAS and CMS collaborations of CERN announced the discovery much awaited Higgs boson in early July 2012 verifying the underlying concept of electroweak symmetry breaking (EWSB) in the SM of particle physics. This triggered essential precise studies of this particle to confirm its nature and to discover some new physics in case of some deviation. With available LHC data, measured properties of Higgs are consistent with SM. Inclusive and differential cross section measurements of Higgs are its very important properties which help to probe its structure, its nature of interaction with decaying particles. These measurements provide a direct probe into SM Higgs sector perturbative QCD calculations at the TeV energy scale.

Measurement of inclusive and differential cross section of the Higgs boson production in the $H \rightarrow 4\ell$ decay channel ($\ell = e, \mu$) in proton-proton collisions at $\sqrt{s} = 13$ TeV corresponding to integrated luminosity of 137.1 fb^{-1} is presented in this dissertation. To minimize the dependence of the measurements on the underlying model of the Higgs boson production mechanism and its properties, the cross sections are extracted in a fiducial phase space closely matching with the experimental acceptance. The remaining model dependence is extracted by repeating the measurement for SM and other exotic model and reported as a distinct systematic uncertainty. All measurements include the correction for the finite detection efficiency and resolution effects. As $\sigma \times BR$ strongly depend upon Higgs mass, the measurement has been extracted assuming $m_H = 125.09 \text{ GeV}$, as measured using the $H \rightarrow 2\gamma$ and $H \rightarrow 4\ell$ decay channels by the CMS experiment [Reference39].

The differential fiducial cross sections are presented using the full Run 2 CMS data collected at 13 TeV as a function several kinematic observables which are sensitive to the Higgs-boson production mechanisms: rapidity and transverse momentum of the four-lepton system, associated jet multiplicity and transverse momentum of the leading jet. Full Run 2 data enabled us to explore the differential observables in more fine bins than before. The integrated fiducial cross sections are measured to be $2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\text{stat.})^{+0.24}_{-0.19}(\text{syst.})$ fb at 13 TeV.

All results have been corrected for the detection efficiency and resolution effects to allow for direct comparison with the SM predictions. The agreement between SM predictions and measured $H \rightarrow 4\ell$ cross section measurement at $s = 13$ TeV is found be good. These model independent cross section measurements can be compared with any theoretical predictions in future with better accuracy.

1.2 Outlook

After the discovery of the Higgs boson, it became very natural and important to measure Higgs boson properties as precise as possible. Precision measurements of the Higgs boson properties can provide us direct or indirect hint of any new physics for which we are looking desperately. This dissertation presented the measurements of integrated and differential Higgs boson via $H \rightarrow 4\ell$ decay channel which is an important Higgs boson property. These measurements provide a direct probe into SM Higgs sector perturbative QCD calculations at the TeV energy scale. Any deviation of integrated or differential distributions from SM expectations would provide a direct hint to BSM contributions. The differential distributions has been extracted only for 13 TeV dataset. These differential distributions are sensitive gluon fusion production mechanism, PDFs of colliding proton, theoretical modeling of hard quark radiation and relative contribution of different Higgs boson production mechanism.

The current measurements are emerged with significant statistical uncertainties. The studies based on simulations showed that the uncertainties on Higgs cross section measurements with 300 pb collisions data would be significantly less than the current theoretical uncertainties. LHC has aim to deliver 3000 of pb collisions data. The Higgs cross section measurements would be extremely important along with other properties with more data.

Bibliography

- [1] F. Englert and R. Brout. “Broken Symmetry and the Mass of Gauge Vector Mesons”. In: *Phys.Lett.* 13.9 (1964), pp. 321–323.
- [2] T. W. B. Kibble. “Symmetry Breaking in Non-Abelian Gauge Theories”. In: *Phys. Rev. Lett.* 155.5 (1966), pp. 1554–1561.
- [3] C. R. Hagen G. S. Guralnik and T. W. B. Kibble. “Global Conservation Laws and Massless Particles”. In: *Phys. Rev. Lett.* 13.20 (1964), pp. 585–587.
- [4] P. W. Higgs. “Broken symmetries, massless particles and gauge fields”. In: *Phys. Rev. Lett.* 12.2 (1964), pp. 132–133.
- [5] P. W. Higgs. “Broken Symmetries and the Masses of Gauge Bosons”. In: *Phys. Rev. Lett.* 13.16 (1964), pp. 508–509.
- [6] P. W. Higgs. “Spontaneous Symmetry Breakdown without Massless Bosons”. In: *Phys. Rev. Lett.* 145.4 (1965), pp. 1156–1163.
- [7] ATLAS Collaboration. “Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC”. In: *Physics Letters B* 716 (2012), pp. 1–29.
- [8] CMS Collaboration. “Observation of a new boson at a mass of 125 GeV with the CMS experiment at the LHC”. In: *Physics Letters B* 716 (2012), pp. 30–61.
- [9] ATLAS Collaboration. “Evidence for the spin-0 nature of the Higgs boson using ATLAS data”. In: *Phys. Lett. B* 726 (2013), pp. 120–144.
- [10] G. Aad et al. “Measurements of fiducial and differential cross sections for Higgs boson production in the diphoton decay channel at $\sqrt{s} = 8$ TeV with ATLAS”. In: *JHEP* 2014.9 (2014), pp. 1–61.
- [11] G. Aad et al. “Measurement of the Higgs boson production cross section at 7, 8 and 13 TeV center-of-mass energies in the channel $H \rightarrow \gamma\gamma$ with the ATLAS detector”. In: *JHEP* 2014.9 (2014), pp. 1–61.

- 106 [12] G. Aad et al. “Fiducial and differential cross sections of Higgs boson produc-
107 tion measured in the four-lepton decay channel in pp collisions at $\sqrt{s} = 8$
108 TeV with the ATLAS detector”. In: *Physics Letters B* 738 (2014), pp. 234–253.
- 109 [13] ATLAS collaboration. “Measurements of the Higgs boson production cross
110 section at 7, 8 and 13 TeV centre-of-mass energies and search for new physics
111 at 13 TeV in the $H \rightarrow ZZ^* \rightarrow \ell^+ \ell^- \ell'^+ \ell'^-$ final state with the ATLAS detector”.
112 In: *ATLAS-CONF-2015-059* (2015).
- 113 [14] G. Aad et al. “Measurements of the Total and Differential Higgs Boson Pro-
114 duction Cross Sections Combining the $H \rightarrow \gamma\gamma$ and $H \rightarrow ZZ^* \rightarrow 4\ell$ Decay
115 Channels at $\sqrt{s} = 8$ TeV with the ATLAS Detector”. In: *Phys. Rev. Lett.* 115
116 (2014), p. 091801.
- 117 [15] ATLAS collaboration. “Measurements of the total cross sections for Higgs
118 boson production combining the $H \rightarrow \gamma\gamma$ and $H \rightarrow ZZ^* \rightarrow 4\ell$ decay chan-
119 nels at 7, 8 and 13 TeV center-of-mass energies with the ATLAS detector”. In:
120 *ATLAS Note ATLAS-CONF-2015-069* (2015), pp. 1–61.
- 121 [16] V. Khachatryan et al. “Measurement of differential cross sections for Higgs
122 boson production in the diphoton decay channel in pp collisions at $\sqrt{s} = 8$
123 TeV”. In: *Eur. Phys. J. C* 76 (2016), pp. 1–31.